Reference, Norms and the Cost of Interaction: A Norm-Setting Model of Autistic–Non-Autistic Communication, with a Discriminating Design

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Abstract

How much of the cost of an interaction between an autistic and a non-autistic person is set by the norms that govern it, and when does changing those norms change, or reverse, which partner pays? The paper starts from a thought experiment, a clinic whose manual is written from an autistic reference, and from the result that makes it exact: comparative deficit verdicts between two groups trade places under a change of reference and of the inventory of competences exactly when the groups stand in mutual advantage at the measurement's margin. The result concerns verdicts, not the structure of the populations or their diagnoses, and it motivates without entailing a cost model: effort is a participant baseline plus a symmetric mismatch between the participant's communicative style and the style the governing norms presuppose, plus a one-sided shortfall against the competence the norms require. The model's predictions follow exactly, among them that the group gap reverses only when the norm-induced reduction exceeds the whole initial gap. A within-dyad crossover is designed in which the same mixed pairs work under two explicit rule sets, with profiles measured and norms coded before any interaction, so that each participant's change in effort is predicted from quantities fixed in advance, against the rivals of a general protocol benefit and partner mismatch. A simulation at the planned size recovers the parameters, predicts the relative individual effects of a withheld rule set for held-out dyads to the noise floor, and reports which conclusions survive measurement error and correlated baselines; the partner term is identified only under an assumption the design cannot enforce.

1 Introduction

1.1 The question

When an autistic and a non-autistic person interact and the interaction goes badly, the received description locates the difficulty in one of them. The double empathy problem [2] relocated it to the pair: a mutual failure of understanding between people whose ways of experiencing and communicating differ, not a one-sided deficit, a relocation with a decade of empirical work behind it [3, 4, 5]. This paper asks about a third party present in every such interaction, the norms that

*Email: akb9408@rit.edu, Website: `https://aethus.org`. This paper develops, as a self-contained proposal in the psychology of communication, two principles stated in a larger framework [1]: the Branch Principle and the Principle of Symmetry, with the observation that a difference which switches with context refutes an unconditional claim of incapacity. The framework's metaphysics plays no role in the argument; what is carried over is one piece of bookkeeping, the treatment of a diagnostic label as a partial information state, explained where it enters.

govern it. Milton's account already has them in view: social subtext is actively constructed, and a label that locates the problem in the other resolves the one who applies it [2]. What the existing accounts do not have is the norms as a measured and manipulable variable, with a prediction of how a change in them redistributes effort between the partners. Any interaction runs under rules about how a request is made, how a turn is yielded, how much is said outright and how much left to be inferred, and those rules presuppose a communicator. The question is how much of the interaction's cost is set by that presupposition, and when changing the rules changes, or reverses, which partner pays it.

The paper starts from a thought experiment, set out in Section 3.1: a clinic whose manual was written from an autistic reference, and its intake notes on one hypothetical client from the majority, in which familiar behaviour reads as symptom because of where the reference stands. The result that makes the experiment exact is that comparative deficit verdicts between two groups trade places under a change of reference and inventory exactly when the groups stand in mutual advantage at the measurement's margin, a condition that can be measured. The result is about verdicts. It does not say that the two groups are not distinct types, and it does not entail the cost model; it motivates the question the rest of the paper asks, whether the labour of a mixed interaction moves when the conventions it runs on move, and conventions can be changed in a laboratory.

The question has three parts, and the paper keeps them apart because they have different standing.

- *A definitional part.* A verdict of deficit is relative to a reference. This is a matter of definition, and it is proved below in the only sense a definition can be proved, by exhibiting the quantifier the verdict hides.
- *A formal part.* Under a cost model with a stated structure, the group gap in effort under given norms, its change under new norms and the condition for its reversal follow exactly, and the reversal condition is stricter than it is usually taken to be.
- *An empirical part.* Whether effort tracks the mismatch between a person and the norms in the way the model says is a hypothesis, and the paper's main work is a design that can tell it from its strongest rivals.

1.2 What the paper adds

The contribution is operational. Interactions between differently situated people have an established theoretical setting in the study of aptitude-by-treatment interactions [6, 7]: who benefits from a condition depends jointly on the person and the condition, and context-dependent advantages and their reversals are at home there. The double-empathy literature has a formal model of its own, a feedback model of channel preference in which each partner's registered empathy depends on the partner's verbal and nonverbal output weighted by the first partner's preference [8]. What this paper adds is narrower than either and not contained in them: a prediction of how an externally imposed communication norm distributes the cost of adaptation between the two partners of a dyad, from profiles measured before the interaction and norms coded independently of its outcome, beyond the effect of who the partner is. The model that carries the prediction is one of a family stated together in Section 6, each rival a parameter restriction, and no tradition is identified with a restriction.

1.3 Where the model comes from

The paper develops two principles the author stated in a larger work [1], in a section on the relation between cognitive style and the conventions of a dominant group; Appendix C gives the passages in full, with their provenance. The *Branch Principle* holds that "ontologically neurodivergent people are better understood as a collection of categorical branches and 'allistic' is a single branch, such that two random branches are typically as different from one another as they each are to a third". The *Principle of Symmetry* holds that "you can always redraw the conventional category boundaries in psychology so that allistic and a given neurodivergent branch 'trade places phenomenologically'". Of what the passages say, the paper separates four things: reference-relativity, which is definitional; reversibility, the "trade places" claim, which is the thought experiment of Section 3.1 and holds exactly under mutual advantage at the measurement's margin; "roughly half-and-half", a conjecture nothing below rests on; and the remark that autistic people "are described as taking a script to allistic conversations", when "if the shoe were on the other foot" allistic people would be, which is the seed of the cost model and the hypothesis the paper is built to test. The Branch Principle, a claim about the geometry of cognitive profiles, is an extension with its own test (Section 10).

Two commitments govern the writing: the scope rule of Section 2 for every claim about "the autistic", and the recommendations to autism researchers on ableist language [9], so that "autistic" and "non-autistic" are the groups' names, "allistic" appears only inside the source's quotations, and no competence difference is called a deficit except relative to a stated reference.

1.4 Plan

Sections 2 to 4 are the preliminaries: a label as a partial information state and the scope rule; the redrawing thought experiment and the Principle of Symmetry made exact; what a difference that switches with context does and does not show. Section 5 gives the cost model and derives its predictions, Section 6 the model family and what can and cannot separate its members, Section 7 the study, Section 8 the simulation with its robustness checks, Section 9 what the published results already constrain, Section 10 what lies outside the paper, and Section 11 the discussion. Appendix A states the simulation's likelihood, Appendix B the preregistration items, and Appendix C the source passages.

2 A label as a partial information state

A diagnostic label carries what it records and no more. The larger framework calls such a state an Aethus and any fuller description of the same person a proper child of it [1]; this paper says *partial information state* and *refinement*. The point of the vocabulary is not metaphysical. It is that a label is an object with a definite content, and that a claim about everyone who bears it is licensed by that content or not at all.

Three kinds of claim about the bearers of a label are therefore distinguished.

(i) *Definitional entailment.* What the label records holds of every bearer. The diagnostic criteria for autism spectrum disorder in the fifth edition of the *Diagnostic and Statistical Manual* require all three of the social-communication features and at least two of the four restricted and repetitive features, currently or by history [10]. So "every bearer shows at least two of the four" is entailed. "Every bearer has sensory differences" is not: sensory reactivity is one of the four, and the label does not record which two.

(ii) *Group-level statistical prediction.* A claim about the distribution of bearers. It is true or false of the group, and attributing it to each member is the ecological fallacy [11].

(iii) *Individual attribution.* A claim about a particular bearer, which needs information about that bearer.

Definition 1 (Scope rule)**.** A claim about every bearer of a label is licensed in two ways only: the label's definition entails it, or independent evidence establishes it for every bearer. A claim that needs information the label does not carry is made of a refinement that carries it, or not made.

The rule is the discipline the rest of the paper is written under. It does not forbid universal claims; it says where they come from.

The source's illustration. The section the principles come from gives one worked instance of a claim that violates the rule (Appendix C): for any trait some autistic branch can be found on which it presents as infantile, which seems to confirm that "autistic people are infantile in this way". Two errors are in play. With $I(b, \tau)$ reading "τ presents as infantile on b relative to a chosen reference branch" and $\mathcal{B}$ the branches the label collects, the category-to-individual step $\exists b \in \mathcal{B}\ I(b, \tau) \not\Rightarrow \forall b \in \mathcal{B}\ I(b, \tau)$ is the scope rule's violation, and the interchange of $\forall \tau\, \exists b\ I(b, \tau)$ with $\exists b \forall \tau\ I(b, \tau)$ is what makes the collected witnesses look like one profile. The source's random-partition control shows that the confirmations depend on the partition chosen, the reference class problem [12] in categorical form, with the qualification that an arbitrary partition need not place a witness for every trait in the half containing the allistic branch.

3 The Principle of Symmetry made exact

3.1 The redrawing: a thought experiment

The illustration is stipulated. Consider a clinic whose reference population is a branch of autistic adults and whose diagnostic manual was written from their norms, and read its intake notes on one hypothetical client from the majority.

> The client seeks sustained eye contact during speech and reports discomfort when it is not returned. Requests are seldom stated; the client encodes them as hints, expects the listener to reconstruct them, and is frustrated when the literal content is acted on. Attention moves between topics within a session, and the client reports discomfort when held to one. In group settings the client engages in repetitive rhythmic whole-body movement to music, and does not experience it as stereotyped.

Four behaviours, each stipulated of this client and none attributed to the majority as a whole, an attribution the scope rule of Section 2 forbids. What the notes do is describe each in the register the majority's manual keeps for the minority: dancing written up as stimming; a breadth of attention as the inverse of monotropism, the attention tunnel of few interests highly aroused [13]; reliance on implicature as a difficulty with literal content. A reader who, with the heading covered, finds the notes plausible as a description of an autistic client has felt the point from the inside, and the point is about the evaluative work the clinical register does. The notes separate four things a diagnosis runs together: the behaviour, its function for the person, its description, and whether it impairs; the register fixes the third and insinuates the fourth, and the thought

experiment shows the third moving while the first two stay put. The source's own first version of the experiment, in 2024, ran the inversion on "logical" against "nuanced" (Appendix C).

The claim the experiment makes exact is the next subsection's: for any two branches there is a redrawing of the categories, a change of reference and of the inventory of competences, under which their *comparative deficit verdicts* trade places, exactly when the two stand in mutual advantage at the measurement's margin, each exceeding the other somewhere by more than the instrument resolves. The condition is a statement about measured profiles, answered competence by competence; the paper's reading of the autistic–non-autistic case is that it is satisfied, which the premeasured competence profiles of Section 7 can check on their inventory, though nothing in the model's test depends on it (Section 3.3). Four claims are then to be kept apart, since the experiment invites running them together.

(i) Selected deficit comparisons reverse when the reference and the inventory change. This is proved, under the stated condition.

(ii) An autistic and a non-autistic person's clinical diagnoses trade places. This needs more than (i): a diagnosis under the criteria the paper cites requires a developmental history and clinically significant impairment beside the behavioural features [10], and rewriting behaviour in clinical language does not establish that those are met. Whether those requirements are themselves reference-relative is a further argument, not made here.

(iii) The diagnostic distinction has no basis in the structure of the populations. This does not follow from (i). Two populations clustered tightly at competence profiles $(10, 1)$ and $(1, 10)$ satisfy the condition, and selecting either competence reverses which appears deficient, while the populations remain sharply distinguishable types with different causes. Reference-relative ranking and the structure of the population are different questions; the second is the Branch Principle's, and it is outside this paper (Section 10). The source compares the two-category picture to the pre-Darwinian division of humans from animals (Appendix C); that comparison belongs to the structural question, not to (i).

(iv) Changing the governing norms redistributes effort. This is an independent empirical hypothesis, the cost model of Section 5. A world with crossed competence advantages and no effort response to a change of norms is consistent with (i).

What the experiment establishes is (i); what it motivates is (iv). The deficit reading of the divide is reference-relative, and the cost of a mixed interaction can therefore be asked about in the relation between a person and the conventions the interaction runs on rather than in the person; conventions can be changed in a laboratory, and the rest of the paper asks that question; the experiment does not answer it.

3.2 Profiles, references and deficit sets

Let T be a finite inventory of competences and let each branch b have a profile $x_b \colon T \to \mathbb{R}$, a score on each competence on a common scale. The scale carries a measurement margin $\delta \geq 0$, the smallest difference the instrument resolves or the analyst is prepared to call a difference.

Definition 2 (Deficit set)**.** The deficit set of branch b relative to a reference branch r at margin δ is

$$D_r(b; \delta) \;=\; \{\, \tau \in T : x_b(\tau) < x_r(\tau) - \delta \,\}.$$

An inventory $S \subseteq T$ shows b in deficit relative to r when $S \subseteq D_r(b; \delta)$ and $S \neq \emptyset$.

Three statements follow, of three different kinds.

(S1) Reference-relativity, a definition. A deficit verdict is a three-place relation among a branch, a reference and a competence, and $D_r(b;\delta)$ has no reference-free extension: there is no set "the deficits of b" until r is fixed. Choosing another reference makes another comparison, not a correction of the first. This is the content of the source's formalization, "the reference-relativity being the crux rather than a modeling convenience".

(S2) Reversibility at margin δ, an exact consequence.

Proposition 1 (Reversibility)**.** *For branches b and r and margin $\delta \geq 0$, the following are equivalent:*
(a) some inventory shows b in deficit relative to r, and some inventory shows r in deficit relative to b;
(b) there are competences τ and σ with $x_b(\tau) - x_r(\tau) > \delta$ and $x_r(\sigma) - x_b(\sigma) > \delta$.
*Call (b) δ-*mutual advantage*.*

Proof. (a) says that $D_r(b;\delta)$ and $D_b(r;\delta)$ are both non-empty. $\sigma \in D_r(b;\delta)$ is $x_r(\sigma) - x_b(\sigma) > \delta$, and $\tau \in D_b(r;\delta)$ is $x_b(\tau) - x_r(\tau) > \delta$, which is (b). □

Under δ-mutual advantage, then, the "trade places" of the Principle of Symmetry (Section 1.3), the redrawing of Section 3.1, holds in this sense and in this sense only, of comparative deficit verdicts: restricting the inventory to $D_r(b;\delta)$ shows b in deficit everywhere, and restricting it to $D_b(r;\delta)$ shows r in deficit everywhere, so that the verdict is reversed by exchanging the reference and the inventory. It holds exactly when the premise holds and fails exactly when it does not. Two remarks fix the premise.

- *Non-domination is not enough above $\delta = 0$.* At $\delta = 0$, (b) is the statement that neither profile is at or above the other on every competence. At a positive margin it is stronger. Take $T = \{\tau_1, \tau_2\}$, $x_b = (1.05, 1)$, $x_r = (1, 1.05)$ and $\delta = 0.1$: neither profile dominates, yet both deficit sets are empty, and no inventory shows either in deficit.

- *What is testable.* Whether a non-reference branch exceeds the reference somewhere by more than the margin is a question about measured profiles, and it is answered competence by competence. The source's own case, two competences at opposite developmental ages (Appendix C), illustrates the premise's shape; it is offered as motivation and not as a measurement on commensurate scales, and it carries no evidential weight here.

(S3) Balance, a conjecture.

Conjecture 1 (Balance)**.** For the reference branch and a given non-reference branch, the deficit sets $D_r(b;\delta)$ and $D_b(r;\delta)$ carry comparable weight under a natural weighting of the competences.

This is the source's "roughly half-and-half". It needs a weighting of competences, and no weighting is natural until the inventory's granularity is settled, since splitting one competence into five multiplies its weight; the paper draws nothing from it.

3.3 Why mutual advantage is not a premise of what follows

The cost model of Section 5 predicts that a change of norms moves effort whether or not the two groups stand in δ-mutual advantage: a participant can gain from a change of norms while scoring lower on every measured competence, if the new norms presuppose less of what that participant lacks, and a participant who exceeds the reference somewhere gains nothing from norms that

do not draw on it. The principles motivate the comparison the model makes; they are not its premises, and the model's test does not test them, which lets the empirical work proceed without settling the geometry of cognitive profiles (Section 10).

4 Context sensitivity: what toggling shows

The source makes one further observation, in a different section and about a different convention, which bears on how any difference between groups should be read: "A limitation does not toggle." An adult who switches between two modes of a convention within a day, by context, shows that "the convention is a mode the platform runs, selectable, not a ceiling the platform imposes" (the passage is in Appendix C). The observation is stated for one person switching between two modes by context, and against one rival, a fixed limitation. Against that rival it is decisive, and the paper takes exactly that much from it.

Proposition 2 (What a toggle refutes)**.** *Let P be a performance and C a set of conditions. If P is observed under some $c \in C$, the claim that the person cannot produce P under any condition in C is false.*

The proposition is immediate, and its value is in what it does not say: persistence does not identify capacity, since a convention can persist across every instrument that shares its assumptions, and change does not identify convention, since a stable capacity yields different performance under different demands. The toggle is a test of *context sensitivity*. A change refutes the unconditional claim of incapacity and opens a question of mechanism, which further experiments answer by varying demand, time, load and format separately from the governing norm. In the study of Section 7 the two conditions differ in the governing norm, are matched on specified incidental features, and have their remaining differences in training and practice measured, which is what lets a change be read as the norm's up to those measured differences.

Applying the observation to differences between groups is this paper's extension, and it is labelled as such. The participant-specific baseline a_i of the model below is accordingly a baseline over the conditions studied and not a measure of capacity, and the components of it that might vary with the conditions, sensory load and fatigue, are held fixed or measured in the design.

5 The cost of interaction under governing norms

5.1 Profiles and norms

Each participant i has two profiles, both measured before any interaction.

- A *style profile* $s_i \in \mathbb{R}^p$: the participant's communicative preferences on dimensions such as directness, literalness, turn-taking and channel, scored on common scales.
- A *competence profile* $k_i \colon T \to \mathbb{R}$ over the abilities the interaction draws on, the profile of Section 3 measured for individuals.

A governing norm n, a rule set under which the interaction runs, fixes two things.

- A *presupposed style* $s_n^* \in \mathbb{R}^p$: the communicator the rules are written for, coded from the rules' text by raters who do not see the outcomes.
- A *competence requirement* $\theta_n \colon T \to \mathbb{R}$ with weights $w_n \colon T \to [0, 1]$: how much of each competence the rules call on, and how heavily.

5.2 The cost model

Hypothesis 1 (The cost model)**.** Participant i's effort under norm n is

$$c_i(n) \;=\; a_i \;+\; m_i(n) \;+\; \varepsilon_i(n), \qquad m_i(n) \;=\; \lambda_s \, d\big(s_i, s_n^*\big) \;+\; \lambda_k \sum_{\tau \in T} w_n(\tau) \, \big[\theta_n(\tau) - k_i(\tau)\big]_+ , \quad (1)$$

with $[u]_+ = \max(0, u)$, d a distance on style profiles, $\lambda_s, \lambda_k \geq 0$, a_i a participant-specific baseline over the conditions studied and $\varepsilon_i(n)$ zero-mean noise.

The two terms of the mismatch $m_i(n)$ have different shapes, and the difference is the point.

- *The style term is symmetric.* Departing from the presupposed style in either direction costs effort: a direct communicator under indirect rules and an indirect one under direct rules both adapt. A distance is the right form for a mismatch of styles, where neither pole is better.
- *The competence term is one-sided.* Falling short of a requirement costs effort; exceeding it costs nothing. A participant whose competence exceeds what the rules call on is not penalized for the excess, which a symmetric distance on competences would wrongly do.
- *The baseline is a baseline.* a_i collects whatever is constant across the conditions studied: capacity, in part, but also sensory load, fatigue and anything else the conditions share. It is not identified as capacity, and the model does not call it that.

Status. Equation (1) is a hypothesis about effort, which the paper tests. Reference-relativity, (S1), makes the comparison between a participant and a norm well defined; the hypothesis says how effort tracks it. Neither the Branch Principle nor reference-relativity entails the equation, and nothing in the framework it comes from does either.

Outcomes. Effort is the primary outcome, measured per participant and per condition by a subjective workload scale, the mental-demand and effort items of the NASA Task Load Index [14] adapted to conversation; one behavioural index fixed in advance is a secondary measure of convergent evidence. Task performance and rapport are secondary outcomes, each with a model of its own; the model does not derive rapport from costs, since "depends on" yields no prediction. Expected and perceived partner empathy are measured in each condition, for the mediation analysis of Section 6.

5.3 The predictions, exactly

Write $G(n)$ for the expected group gap in effort under norm n, the minority group's mean less the majority group's,

$$G(n) \;=\; \Delta a \;+\; \Delta M(n), \quad (2)$$

with Δa the baseline gap and $\Delta M(n)$ the gap in mean mismatch, both signed in the same direction.

P1, the gap under the default norms n_0**.** $G(n_0) = \Delta a + \Delta M(n_0)$. Under norms that presuppose the majority's style and draw on its competences, $\Delta M(n_0) > 0$ whenever the minority's profiles lie farther from $s_{n_0}^*$ and fall shorter of θ_{n_0}, and the gap is the sum of two parts that the default condition cannot separate.

P2, norm sensitivity, the primary prediction. For a second norm n_1,

$$G(n_1) - G(n_0) \;=\; \Delta M(n_1) - \Delta M(n_0), \tag{3}$$

whatever Δa is; and within each person the baseline cancels,

$$c_i(n_1) - c_i(n_0) \;=\; m_i(n_1) - m_i(n_0) \;+\; \varepsilon_i(n_1) - \varepsilon_i(n_0), \tag{4}$$

so that a participant's change in effort is predicted, up to noise, by the change in a quantity measured before the interaction. Equation (4) is the strongest test the model offers, because it runs on premeasured profiles, person by person, and because the restrictions M_γ and M_μ of Section 6 predict a slope of zero for it.

P3, reversal, the strong form.

Proposition 3 (The reversal condition)**.** *Suppose $G(n_0) > 0$. The gap reverses under n_1, $G(n_1) < 0$, exactly when either of the following equivalent conditions holds:*

(a) the norm-induced reduction exceeds the initial total gap, $\Delta M(n_0) - \Delta M(n_1) > G(n_0)$;

(b) under the new norms the mismatch term favours the minority by more than the baseline gap, $-\Delta M(n_1) > \Delta a$.

Proof. By (2), $G(n_1) = G(n_0) - [\Delta M(n_0) - \Delta M(n_1)]$, which is negative exactly under (a); and $G(n_1) = \Delta a + \Delta M(n_1)$, which is negative exactly under (b). □

The condition is stricter than the natural reading of "the norm effect outweighs the baseline". Take $\Delta a = 3$ and let the mismatch contribution move from $+2$ under n_0 to -2 under n_1. The reduction is 4, larger than the baseline gap, but smaller than $G(n_0) = 5$: the gap falls from 5 to 1 and does not reverse. Reversal needs a reduction greater than 5, equivalently a new mismatch contribution below -3; Figure 1 draws the arithmetic. In the language of aptitude-by-treatment interaction, P2 is an interaction of group with condition, and P3 is the condition for it to be disordinal, the groups' lines crossing between n_0 and n_1.

6 The model family, and the test that separates its members

6.1 One model, four restrictions

Add to (1) the two terms a rival would supply: a condition effect γ_n common to all participants, and a mismatch with the partner. For participant i with partner j under norm n,

$$c_i(n, j) \;=\; a_i \;+\; \gamma_n \;+\; m_i(n) \;+\; \mu\, d\big(s_i, s_j\big) \;+\; \varepsilon_i(n). \tag{5}$$

The models compared are restrictions of (5), and each makes a definite prediction for a change of rule set with the partners held fixed (Table 1).

No tradition is identified with a restriction. An account that emphasizes individual limitations can predict effects of task demands and of accommodations, and so is not M_γ or M_μ by fiat; a double-empathy account can include norm effects, and so is not M_μ by fiat. The restrictions are compared as models, by what they predict for the data of Section 7.

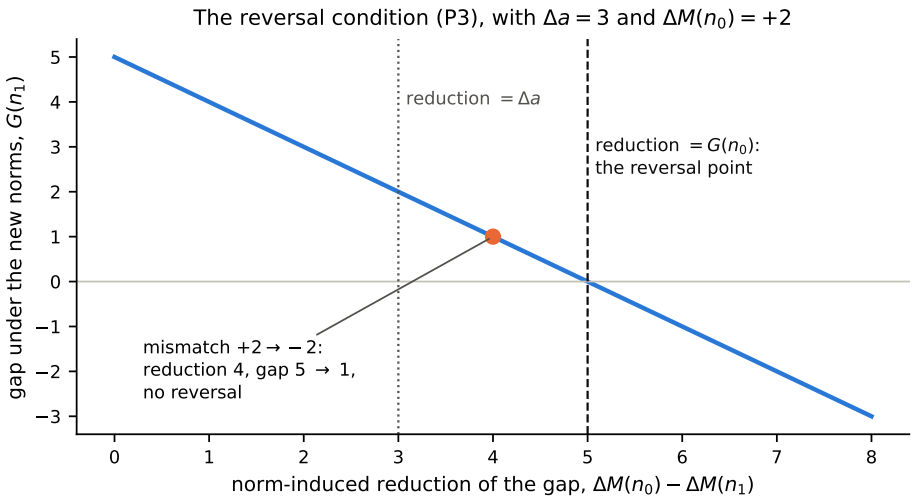


Figure 1: The reversal condition of Proposition 3. The gap under the new norms falls linearly in the norm-induced reduction; it closes when the reduction equals the whole initial gap $G(n_0)$, not when it equals the baseline gap Δa. The marked point is the worked case in the text: a mismatch contribution moving from $+2$ to -2 reduces the gap by 4, more than the baseline gap of 3, and the gap falls from 5 to 1 without reversing.

6.2 The double-empathy model, read from its text

The formal model the double-empathy literature has is the distinct communication strategies model [8], and its characterization here is taken from the paper's own text rather than from its abstract.

- *The implemented mechanism* is channel preference. Each agent registers the partner's empathy as a weighted average of the partner's verbal and nonverbal output, with the verbal weight the agent's own preference *cp*; the gap between the empathy the agent expects and the empathy it registers raises its defensiveness, which lowers its own output, and the loop can collapse.

- *What the text allows.* It states that expectations "may depend on context, relationship history, communication norms", and holds expected empathy at one value for both agents in its simulations. It reports that "the trajectory to collapse shows greater experiential asymmetry", and predicts, as a falsification criterion, that "the partner with higher *cp* should report lower perceived empathy". Its limitations section says that "the costs are therefore bilateral but need not be equal or arise from the same source".

- *What it has no parameter for.* An imposed or governing norm. Its intervention prediction concerns training that increases verbal empathy expression under stress.

It would therefore be wrong to say that this model predicts equal costs, or invariance under a change of norms; it is silent on norms as implemented, and open to them in its text. The distinction between it and the norm-setting model has to be shown in data, and the data can show it only under assumptions, which are stated next.

Table 1: The model family: four restrictions of (5), and what each predicts when the governing rule set changes and the partners stay the same.

Model	Restriction	Prediction under a change of rule set, partners fixed
M_γ, general protocol benefit	$\lambda_s = \lambda_k = \mu = 0$	Effort moves by γ for everyone; nothing is redistributed, and nothing is moderated by premeasured mismatch.
M_μ, partner mismatch	$\lambda_s = \lambda_k = 0$	Beyond γ, nothing changes, since the partner is unchanged.
M_λ, norm mismatch	$\mu = 0$	Effort shifts toward whoever mismatches the governing rules, in proportion to premeasured mismatch (P2), and reverses only under the condition of Proposition 3.
M_{full}	none	Both the partner and the norm terms.

The mediation analysis, and what it can establish. Suppose the double-empathy mechanism is extended so that expected empathy depends on the governing norm. A change of rule set then acts on effort *through* the gap between the empathy a participant expects and the empathy the participant perceives: the gap moves, and effort moves with it. Under M_λ effort moves with premeasured profile-to-norm mismatch whether or not the gap moves. Three things follow for the design, and together they are less than a test.

(i) *The mediator is the gap, not perceived empathy.* If under rule set A a participant expects 8 and perceives 6, and under rule set B expects 6 and perceives 6, the gap falls from 2 to 0 and the whole change in effort may run through it while perceived empathy has not moved; an analysis that controlled perceived empathy alone would count that change as a norm effect beside the mechanism. The design therefore measures both, expected partner empathy before each condition's task and perceived partner empathy after it, and the mediator is their difference.

(ii) *Controlling the mediator does not identify the direct effect by design.* The rule set is randomized; the gap is not. The coefficient of premeasured mismatch in the within-person regression, with the change in the gap controlled, is the direct effect only if the gap's change is unconfounded with the effort change given the rule set and the premeasured profiles, the sequential ignorability of causal mediation analysis [15, 16], and measurement error in the empathy reports, common causes of the reports and of effort, and feedback from effort to the judgement of the partner each violate it. The analysis is therefore run as an assumption-dependent investigation: direct and indirect effects under sequential ignorability; the sensitivity analysis of Imai, Keele and Yamamoto [16], with the range of confounding judged plausible stated in advance on substantive grounds, not from the measures' reliabilities; and measurement error carried with the confounding by the joint sensitivity analysis of Liu and Wang [17], since error in a difference of two reports, and then in its change, moves the direct and the indirect coefficients in directions a single attenuation factor does not capture. The reports' reliabilities, estimated in the pilot, are inputs to that analysis and not a correction on their own.

(iii) *The two accounts are not exclusive.* Equation (1) does not say by what route mismatch

produces effort, and a positive λ may describe an effect that runs in part through the gap. A direct effect that shrinks when the gap is controlled is consistent with the cost model; no categorical prediction, zero, the full slope or something between, is made for the residual coefficient; and a direct effect that vanishes at a stated precision would show that the first study cannot find a norm effect beside the gap, not that none exists.

What the first study can establish is whether the mismatch effect is present with the gap controlled, and how much of it the gap carries under the stated assumptions and across the stated sensitivity range. What would separate the mechanisms is a study that manipulates expected empathy directly, by disclosure or by instruction, which is deferred (Section 7).

6.3 The established setting

Context-dependent advantages that reverse across conditions are aptitude-by-treatment interactions [6, 7], whose disordinal form is the case in which no condition is best for everyone. The norm-setting model is an instance with two specifications: the aptitude is a premeasured profile, not a trait inferred from the outcome, and the treatment is a norm imposed on both partners at once, so that the interaction of interest is between the norm and the *distribution* of cost within the pair. Within that setting the paper's claim is the one of Section 1: that the distribution is estimable, and separately from the partner effect under an assumption stated in Section 8, by the design that follows.

7 Study 1: a discriminating design

The design manipulates the governing rules, holds fixed what can be held fixed of everything else, and measures what cannot. A comparison of "default conversation" with "an explicit protocol" would change norms, structure, predictability, memory demands and perhaps modality at once, and an improvement under the protocol could be any of them. The study therefore compares two explicit rule sets.

Participants. Mixed dyads, one autistic and one non-autistic adult, with diagnosis by report and the autistic participants' self-identification recorded. Same-neurotype dyads are a later study (see "Deferred" below).

Before any interaction.

(i) Each participant's style profile s_i and competence profile k_i are measured with instruments fixed in the preregistration [18]: for style, self-report and a brief performance measure of directness, literalness and turn-yielding preference; for competence, a task of implicature comprehension and a task of following explicit structure, each scored on the scale the requirement is written in.

(ii) Independent raters, blind to the outcomes, code each rule set's presupposed style s_n^* and competence requirement θ_n from its text, on the same scales.

(iii) The distance d, the weights w_n, the outcome measures and the analysis are fixed in the preregistration. Profiles, metric and norm codings chosen after the outcomes are seen would make the model too flexible to fail.

(iv) The common metric is established, not assumed. Independent coding protects the norm locations from being chosen to fit the outcomes; it does not make a participant's preference score and a coded norm location commensurable. The pilot therefore codes the rule sets on the anchored scales the profile instruments report on, infers the competence thresholds from the rule text on the scale the competence tasks are scored in, and checks the calibration against behaviour: participants' adherence to each rule set and their reported difficulty with it should order with the coded distance, and where they do not the scales are revised before the main study.

The manipulation. Two explicit rule sets differ in the communicator they presuppose and are matched on specified incidental features: length, number of rules, channel, time, and the structure of the task they govern. They are not matched in everything, and the model does not ask them to be. The competence term says that the rule sets draw on different competences; inferring a hint and yielding a turn on an implicit cue make demands that stating a request and yielding in words do not, and those are the differences the model is about. What remains, the learning burden of each rule set and its automaticity after practice, is measured rather than assumed equal: training time to criterion, practice errors and reported effort during practice are recorded for each rule set and serve as the check on the matching, which is treated as a claim the pilot supports or revises. They do not enter the primary analysis as covariates: measured after the participant has met the assigned rules, they can carry part of the adaptation cost the analysis estimates, and adjusting for them would remove that pathway or bias the estimate, the hazard of conditioning on a variable the manipulation affects [19]. The primary estimand is the total condition-dependent mismatch effect; how training burden relates to effort is a separate analysis with its own estimand, stated in the preregistration.

- *Rule set A* encodes indirect, implicature-rich conventions: requests made by hinting, an opening exchange of small talk, turns yielded by implicit cues.
- *Rule set B* encodes direct, literal conventions: requests stated outright, no obligatory small talk, turns yielded in words.

Both are practised to a criterion before the task; adherence to the rules, coded from recordings, is the manipulation check. Which conventions rule set B should encode, and from what sources, is a decision for the preregistration, informed by autistic participants' own accounts of their communication and by the published descriptions of autistic communication preferences; the design does not presume that rule set B is the autistic participants' preferred style, it measures where each participant stands relative to each rule set.

Within dyads. Each pair works under both rule sets, in counterbalanced order, on two parallel versions of a collaborative task, also counterbalanced. The order is assigned to dyads at random and balanced; a period effect is estimated as the order-by-condition term; differential carryover, which a two-period crossover cannot separate from the order effect [20], is guarded against by the parallel task versions and a washout between conditions and is assessed by the adherence coding of the second condition, in which the first rule set's conventions would show if they persisted. The room, lighting and noise are identical across conditions; fatigue is measured after each condition. The pair is its own control: the baseline a_i, the partner term $\mu\, d(s_i, s_j)$ and the dyad's shared conditions cancel in the within-person difference (4).

Outcomes. The primary outcome is one measure, fixed in advance: the subjective workload score, the mental-demand and effort items of the NASA Task Load Index [14] reworded to the conversation just held (how mentally demanding the conversation was; how hard the participant had to work to follow the rules and accomplish the task), on the index's scale, per participant and condition. A behavioural index, the rate of repair initiations and restarts per minute of the participant's speech coded from the recording, is a secondary measure of convergent evidence, not a second primary, so the testing rule has one outcome. Its relation to effort is not assumed: a participant may ask more clarifying questions because that is an effective way to reduce uncertainty, a rule set may make clarification easier or more acceptable, and effortful speech can be fluent, so the pilot establishes what the rate captures, against the workload score and against performance, and whether the per-minute denominator stays comparable when the protocols themselves change speaking time. The raters coding it are blind to the hypotheses, to the participants' group and profiles, and to the condition labels; they are not blind to the conditions themselves, since hints, small talk and spoken turn-yielding are audible, and the design says so rather than claiming a blinding it cannot have. Low reported effort can mean easier coordination or lower engagement, so effort is read with task performance and adherence beside it, and a condition in which all three fall is read as consistent with disengagement rather than with ease. Task performance and rapport, secondary; expected and perceived partner empathy per condition, whose difference is the mediator; rule adherence, the manipulation check; training time, practice errors and practice effort per rule set, the matching check.

Predictions.

- *Primary (P2).* On effort, the rule set interacts with premeasured mismatch within persons: the change in effort from A to B is predicted by the change in m_i, with slope one in the model's units and positive in any units. The group-level redistribution of effort is this prediction's aggregate.
- M_γ. A main effect of rule set only, with no moderation by premeasured mismatch.
- M_μ. No change beyond the main effect, the partner being unchanged.
- *A norm-dependent double-empathy mechanism.* A change carried by the expected–perceived empathy gap. What the first study can say about it is stated in Section 6; it is not a prediction the primary analysis tests.

Analysis. The primary analysis is the within-person difference regression of (4), the change in the workload score on the change in premeasured style mismatch and competence shortfall, with standard errors clustered by dyad. Two questions are kept apart: whether premeasured mismatch moderates the change in effort at all, and whether it does so in the direction the model predicts. The joint Wald test of λ_s and λ_k at the 5% level answers the first; it rejects as readily for a negative coefficient as for a positive one. The preregistered decision rule for *primary success* has three parts, each with its uncertainty: the joint test rejects; at least one coefficient is positive by its one-sided 5% test; and neither coefficient contradicts its sign restriction, a contradiction being the one-sided 5% test of $\lambda \geq 0$ rejecting. The third part is stated with a test and not with the sign of the estimate because the model permits a coefficient of exactly zero, whose estimate falls below zero in half of all studies; a rule that required both estimates nonnegative would fail about half the time in a world with one real effect and one null, and the simulation reports that boundary case. Joint rejection with a sign contradiction is recorded as moderation against the model's direction,

which counts against the cost model as stated and is not success; joint rejection with neither coefficient positive by its one-sided test is recorded as moderation of undetermined direction. The simulation's power is computed for the rule. Evidence for each coefficient separately is secondary, tested one-sided in the predicted direction with the familywise rate held at 5%. Levels are analysed in the actor–partner interdependence model [21], participants nested in dyads and conditions within participants, which identifies μ from between-dyad variation under the assumption stated in Section 8. The four models are compared by their Bayesian information criterion on the exact likelihood of Appendix A, and the mediation analysis is as stated in Section 6. Appendix B lists what the preregistration fixes.

What would count against the model. Stated in advance, with the preregistration, and each with the precision it needs, since a nonsignificant estimate establishes no absence:

(i) a manipulation that changes adherence but moves no one's effort, the condition effect and both mismatch coefficients bounded within their smallest effects of interest;

(ii) a change of effort that premeasured mismatch does not moderate: the joint test not rejecting, and an equivalence test [22, 23] rejecting, for each coefficient, every value beyond its smallest effect of interest; a nonsignificant moderation whose interval still covers the smallest effect of interest is inconclusive, not a disconfirmation;

(iii) a direct effect of premeasured mismatch bounded within the smallest effect of interest once the change in the empathy gap is controlled, with the sensitivity analysis showing the bound robust across the prespecified range of mediator–outcome confounding.

None of these falsifies the model family, whose parameter space contains $\lambda_s = \lambda_k = 0$; what the first two establish is that mismatch effects of the prespecified size are absent for the studied participants and protocols, which is the result that matters. The smallest effects of interest are provisionally a quarter of the simulated values, 0.15 for λ_s and 0.20 for λ_k in the model's units, the sizes at which the simulation's power is reported; that is a sensitivity setting and not a substantive bound, and before the preregistration they are translated, on the fixed measurement scales, into the smallest change in the workload score per unit of premeasured mismatch that would matter, with the pilot's estimates where there are any. Power to detect those effects and power to establish their absence are separate questions, and the sample-size assessment of Section 8 reports both. The third condition would show that the first study cannot find a norm effect beside the double-empathy mechanism, and would send the question to the study that manipulates expectations.

Deferred. Composition, all-autistic and all-non-autistic dyads beside mixed ones, and disclosure of diagnosis to the partner are each a manipulation of their own, and each goes to its own study. Composition tests whether the partner term μ and the norm terms interact, and whether the same-neurotype advantage the first diffusion-chain study found [24], and its replication did not [25], appears under a change of governing norm; disclosure manipulates expectations directly and is the study the mediation analysis of Section 6 defers to. A third design, each participant interacting with several partners in a round robin [26], identifies the partner term within persons rather than under the between-dyad assumption of Section 8, and is the stronger design for μ. A first study that crossed norms with composition, disclosure or repeated partners would need several times the dyads and would confound its own manipulation checks.

Table 2: The simulation's settings. Style and competence profiles are two-dimensional, drawn around group means that overlap; rule sets A and B presuppose opposite styles and draw on different competences; rule set C is withheld from fitting. Effort is in standard-deviation units of the baseline.

Quantity	Setting	Role
Dyads	60 mixed, each under A and B	the planned size; 20 to 120 in the power study
Style profiles s_i	$\mathcal{N}(\pm(0.8, 0.8),\ 0.7^2 I)$	autistic +, non-autistic −, on (directness, literalness)
Competence profiles k_i	$\mathcal{N}(\pm(-0.3, 0.3),\ I)$	autistic +, non-autistic −, on (implicature reading, explicit structure)
Rule set A	$s^* = (-1, -1)$, $\theta = (1, 0)$, $w = (1, 0)$	indirect; requires implicature reading
Rule set B	$s^* = (+1, +1)$, $\theta = (0, 1)$, $w = (0, 1)$	direct; requires following explicit structure
Rule set C	$s^* = (+1, -1)$, $\theta = (0.5, 0.5)$, $w = (0.5, 0.5)$	withheld; direct but non-literal
$\lambda_s, \lambda_k, \mu, \gamma_B$	0.6, 0.8, 0.4, −0.3	the generating values; $\gamma_A = 0$
Δa	0.5	the autistic group's baseline shift
$\sigma_a, \sigma_u, \sigma_\varepsilon$	1.0, 0.4, 0.8	baseline, dyad and residual standard deviations

8 The simulation

Before any data, the design was run on simulated participants to settle five questions: whether the analysis recovers the parameters of (5) at the planned size; whether it selects the generating model among the four; whether a model fitted to two rule sets predicts a third, for participants it has not seen; what the reversal arithmetic looks like at realistic profiles; and, since each of these answers is conditional on the generating model, which of them survive departures from it. Showing that the equations generate the intended pattern would establish much less than this, since any model generates its own pattern. The simulation is the supplementary script `norm_cost_sim.py`, version 1.4, run with the fixed seed 20261023, whose complete output and figures accompany the paper; its settings are in Table 2.

Estimation. The simulation generates effort from (5) with one addition, a dyad effect shared by the two partners, which the within-person difference cancels. The balanced crossover decomposes exactly into within-person differences, which identify γ and the two λ's with the baseline, the dyad effect and the partner term cancelled, and person means, which identify μ beside a group indicator carrying the baseline gap. The second identification is conditional in a way the first is not. With one partner per participant, $a_i' = a_i + \mu\, d(s_i, s_j)$ with $\mu' = 0$ gives the same observations as a_i with μ, so the person means identify μ only under a restriction on how baselines relate to partner distance: the assumption the between-dyad estimate needs is the one the regression actually fitted requires, that the baselines have zero conditional mean given the regressors of (7), the group indicator, the two mean mismatch terms and partner distance. That is stronger than mean-independence of partner distance given the full profiles, since the regression carries the profiles only through the mean mismatch terms: if baselines vary with the profiles' extremity and extremity predicts partner distance, the dependence passes to $\hat{\mu}$ unless the extremity is modelled.

The preregistration therefore adds the measured profile coordinates to the between equation as covariates, and the simulation fits them throughout, so that the power reported for μ is the planned analysis's. Linear coordinate terms absorb linear dependence of the baselines on the coordinates and nothing else: a dependence on the profiles' extremity, which is nonlinear in them, passes to $\hat{\mu}$ in part, and a dependence on partner distance itself passes entirely. The remaining assumption is one the actor–partner model states as a random-effects assumption and the design does not supply, and the robustness study below shows what its failure does; the simulation satisfies it by construction except in the scenario built to violate it. Each part is fitted by least squares with dyad-clustered standard errors, the primary test is the joint Wald test of the two λ's on the clustered covariance, and the model comparison uses the exact likelihood of Appendix A, in which the dyad effect's covariance between partners is carried by a rotation of the two partners' means.

Parameter recovery, and the power of the decision rule. Over 1,000 simulated studies of 60 dyads, every parameter is recovered without bias and with coverage close to nominal (Table 3, Figure 2). The difference equation identifies three parameters within persons, and not equally well: the standard deviations of $\hat{\lambda}_s$ and $\hat{\lambda}_k$ are a tenth of their values, while γ, a difference of condition means against the residual noise, has power 81%; the partner term μ, identified between dyads with the profile coordinates as covariates, has 76%. The power study (Figure 3) is computed for the three-part decision rule of Section 7. At the generating effects the rule succeeds in every replication from 20 dyads, which says only that those effects are large. At a quarter of them, $\lambda_s = 0.15$ and $\lambda_k = 0.20$, the rule has power 60% at 20 dyads, 92% at 40 and 98.5% at 60, and with the profiles and the codings measured with error 58%, 85% and 94%: the planned 60 dyads hold 90% power down to a quarter of the simulated mismatch effects under measurement error, which is the figure the preregistration should quote. At the boundary of the parameter space, one coefficient at its generating value and the other exactly zero, the rule succeeds in 95% and 94% of studies in the two arrangements, losing only the 5% in which the null coefficient's one-sided test contradicts its sign by chance, where a rule requiring both estimates nonnegative would succeed in 49% and 52%; at a quarter of the effects the boundary rates are 66% and 57%. When the truth is M_γ the rule's false-positive rate is 1.9%, below the joint test's 4.9% because a rejection with no coefficient positive by its one-sided test, or with a sign contradiction, is not counted as success; the separate two-sided tests are at 5.1% and 5.3%, and "either coefficient" at 9.8% uncorrected, which is why the separate tests are secondary and familywise-controlled. Power to establish the absence of the effects is another matter. The two equivalence tests at the smallest effects of interest, both 90% intervals inside $(-0.15, 0.15)$ and $(-0.20, 0.20)$, succeed when the truth is M_γ in 38% of studies at 60 dyads (37% over the 1,000 replications of the false-positive study), 71% at 90 and 88% at 120: a study sized to detect the effects at 60 dyads is not sized to rule them out, and a preregistration that advertises the disconfirmation needs the larger sample. γ passes 80% at 60 dyads, μ 90% at 90.

Model recovery. With 500 studies generated from each model and the four fitted to each, the Bayesian information criterion on the exact likelihood selects the generating model in 97% of the M_γ studies, 76% of M_μ, 96% of M_λ and 79% of M_{full} (Figure 4, left); the two large errors both drop the partner term, the parameter the design identifies least well, and the norm terms are almost never mistaken, no study generated with them selecting a model without them and 5 of 1,000 generated without them selecting a model with them. A predictive log score with whole dyads held out, six folds, is reported beside the criterion (right) as the comparison that needs no assumption about the effective sample size; it never drops the norm terms when they are present

Table 3: Parameter recovery over 1,000 simulated studies of 60 mixed dyads, the full model fitted: the generating value, the mean estimate, its standard deviation, the coverage of the dyad-clustered 95% interval, and the power of the two-sided 5% test against zero. The preregistered decision rule of Section 7 succeeds in 100% of these studies and in 1.9% when the truth is M_γ. The levels regression carries the four measured profile coordinates as covariates, as the planned analysis does.

Parameter	Truth	Mean estimate	SD	95% coverage	Power
γ (condition effect)	-0.300	-0.296	0.102	95.1%	81.1%
λ_s (style mismatch)	0.600	0.601	0.057	94.4%	100%
λ_k (competence shortfall)	0.800	0.802	0.085	93.5%	100%
μ (partner mismatch)	0.400	0.403	0.151	94.7%	75.8%

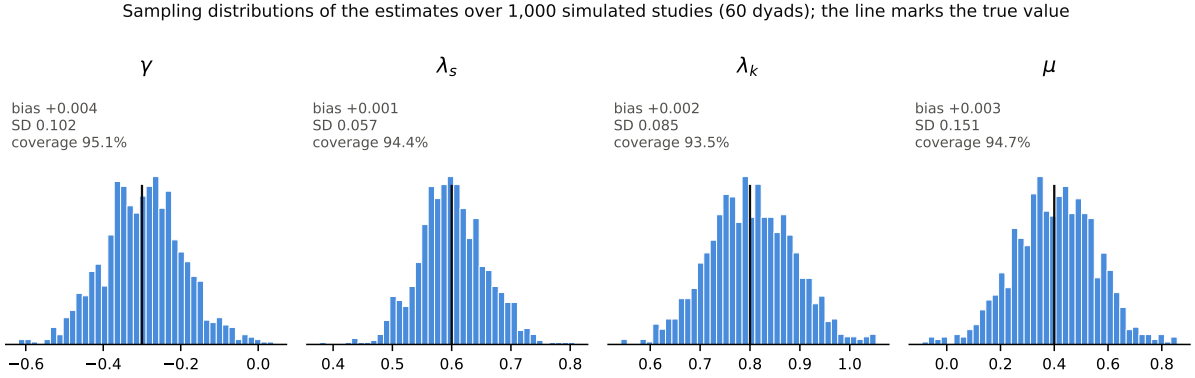


Figure 2: Sampling distributions of the four estimates over 1,000 simulated studies of 60 dyads, the full model fitted. The vertical line marks the generating value. The mismatch parameters λ_s and λ_k are estimated with a precision that γ, identified within persons as a difference of condition means, and μ, identified between dyads, do not reach at this size.

but over-selects when the truth is small, choosing a model with the norm terms in 37% of the M_γ studies and 40% of the M_μ studies, the known behaviour of such scores [27, 28]. The preregistered criterion is therefore the Bayesian one, and whether the norm terms are present at all is settled by the primary analysis, not by the model comparison.

The P2 signature. Figure 5 shows one simulated study under each of two truths. Under norm mismatch the within-person change in effort from A to B falls on a line through the change in premeasured mismatch, with the autistic participants, whose profiles lie nearer rule set B, moving down and the non-autistic participants moving up; under a general protocol benefit the same plot is flat, both groups moving together by γ. This is the figure the real study will produce. Its two premeasured mismatches are combined into one score with the generating weights, fixed before the data; in the real study the weights are fixed in advance from the pilot or estimated on dyads other than those plotted, since a plot whose weights are fitted to the effort data it displays shows the fit and not a prediction, whatever its slope.

The withheld rule set. The mismatch coefficients were fitted to rule sets A and B on five sixths of the dyads and used to predict, for the participants of the held-out sixth, the change in effort from A to the withheld rule set C from their premeasured profiles alone, over six folds.

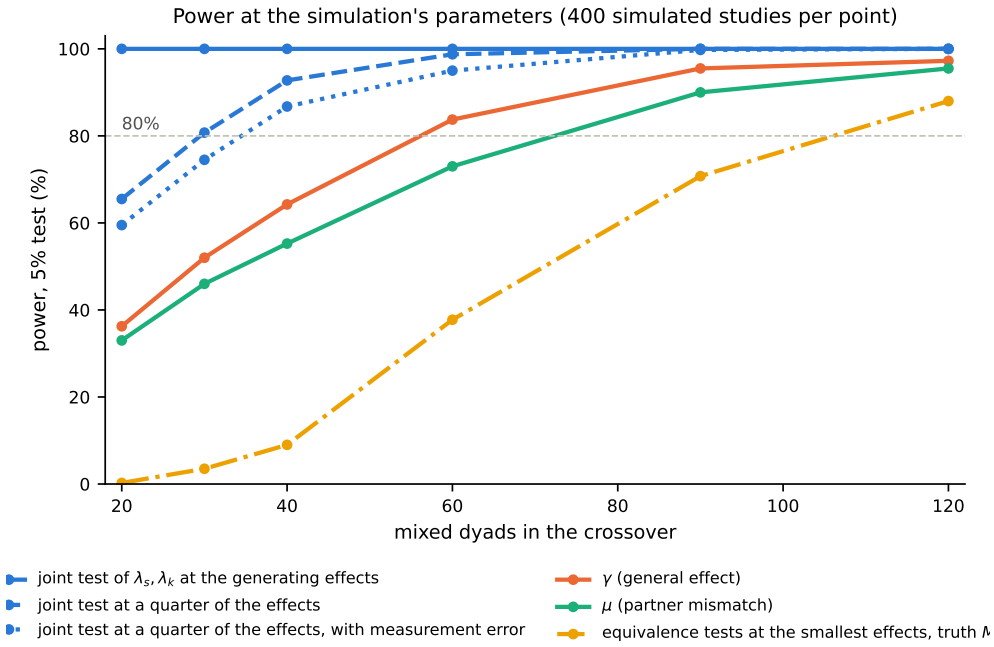


Figure 3: Power against the number of mixed dyads, 400 simulated studies per point. Solid lines: the joint test of (λ_s, λ_k) at the generating effects, and the two-sided tests of the condition effect γ and the partner term μ. Dashed: the joint test at a quarter of the mismatch effects ($\lambda_s = 0.15$, $\lambda_k = 0.20$); dotted: the same with the profiles and the rule-set codings measured with error; the decision rule's power runs within six points of the joint test's at 20 dyads, within three at 30 and within two from 40. Dash-dotted: the power of the two equivalence tests at the smallest effects of interest when the truth is M_γ, which passes 80% only above 100 dyads.

What is predicted is the relative individual change: A and B cannot identify the condition effect γ_C, so the predicted and the observed changes are each centred within the held-out set, and the exercise tests who will find C harder than A, and by how much relative to the others, not the absolute effect of C. With the truth M_λ the root-mean-square error is 1.10 (standard deviation 0.07 over 500 studies) against 1.47 for the rivals' prediction of no individual differences, and the irreducible error of a centred within-person change is $\sqrt{2}\,\sigma_\varepsilon\sqrt{1 - 1/20} = 1.10$ at this fold size: the norm models predict the relative effects of a rule set they have not seen, for participants they have not seen, to the noise floor, and the same holds with the truth M_{full} (1.11 against 1.47). The floor is set by the two observations in the observed change; a replicated baseline measurement would lower it, and the real study's third rule set, if one is run, is where that replication belongs.

The reversal arithmetic. At the generating values, on a population of 20,000 dyads, the mismatch gap ΔM is $+1.45$ under rule set A and -1.44 under B, so that with the baseline gap of 0.50 the total gap G is $+1.95$ under A and -0.94 under B: the norm-induced reduction, 2.89, exceeds $G(\text{A}) = 1.95$, and the gap reverses, as Proposition 3 requires. Under the withheld rule set C, which presupposes a style equidistant from both groups' means, ΔM is -0.003 and the gap is the baseline alone. One population shows the three regimes the proposition distinguishes: a gap set by the norms, a gap reversed by them, and a gap the norms leave to the baseline.

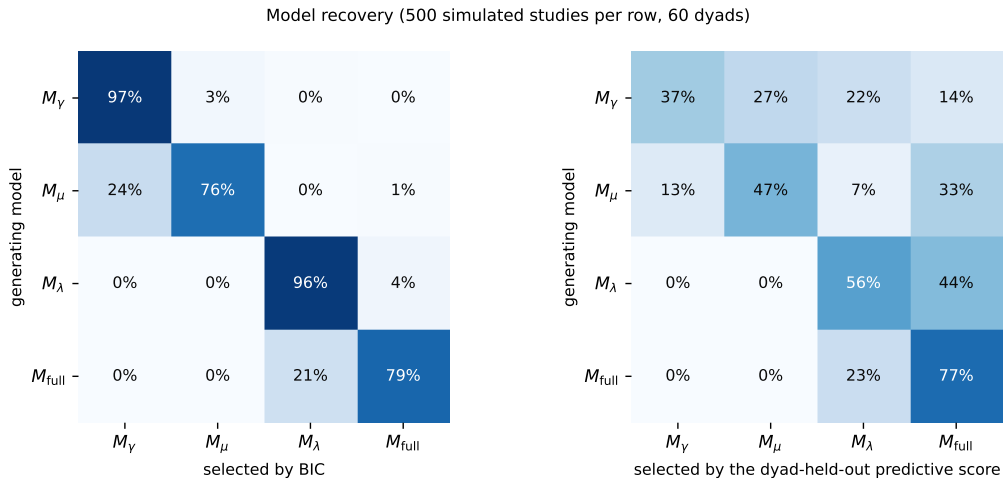


Figure 4: Model recovery: the share of simulated studies (500 per generating model, 60 dyads) in which each of the four models was selected, by the Bayesian information criterion on the exact likelihood (left) and by the dyad-held-out predictive log score (right). Under the criterion the off-diagonal mass lies almost entirely in dropping the partner term; the predictive score over-selects the larger models when the truth is small.

Robustness. Every result above is conditional on the generating model, and the real study will not satisfy it. Table 4 repeats the recovery under five departures from it, singly and together, at 60 dyads. Three conclusions survive all of them: the joint test keeps its power, the Bayesian information criterion keeps selecting a model with the norm terms, and $\hat{\lambda}_s$ and $\hat{\lambda}_k$ stay near their values. Two do not. Measurement error in the profiles and the codings attenuates $\hat{\lambda}_k$ by an eighth and spoils the intervals: a rule set's coding error is one draw shared by every participant, so it does not average out within a study, and the coverage of $\hat{\gamma}$'s interval, which absorbs it, falls to 55%, that of $\hat{\lambda}_k$'s to 74%; the real study's intervals are honest only if the coding error is small or is propagated, by coding each rule set with several raters and carrying their disagreement into the standard errors. And baselines correlated with partner distance and with the profiles' extremity bias $\hat{\mu}$ by more than its value, with coverage 9%, the profile covariates absorbing almost none of it since the extremity is nonlinear in the coordinates and the partner-distance term is not theirs to absorb, as the identification argument above says; the within-person estimates are untouched.

What the simulation does not show. It does not show that the effect sizes are realistic; they are assumptions, and the quarter-effect power curves are the nearest it comes to a guard against them. It does not show that style and competence can be measured on scales commensurate with the codings of the rule sets, which is the pilot's problem. And it does not model the empathy gap, whose analysis depends on assumptions a simulation that built them in could not test.

9 What existing results constrain

Three bodies of results bear on the design, and each is read from its primary source.

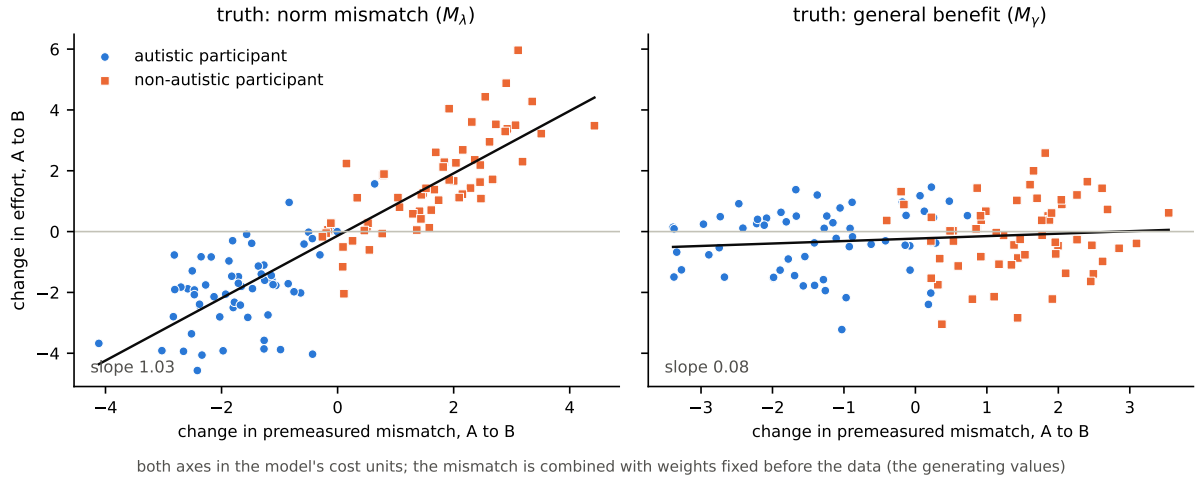


Figure 5: The P2 signature in one simulated study of 60 dyads under each of two truths. Each point is a participant: the change in effort from rule set A to rule set B against the change in premeasured mismatch, combined with weights fixed before the data, in the model's cost units. Under norm mismatch (left) the change lies on a line of slope near one and the groups move in opposite directions; under a general protocol benefit (right) the slope is near zero and the groups move together.

Information transfer in chains. The first diffusion-chain study found that information passed along chains of autistic adults, and along chains of non-autistic adults, more effectively than along mixed chains [24]. The registered replication with 311 participants did not find that difference: there was no main effect of chain type on information transfer, for fictional or for factual stories, and an unpredicted interaction between autistic chain type and position for fictional stories, performance falling more steeply along autistic chains than non-autistic ones [25]. Rapport, self-rated and analysed by role, was higher for teachers in non-autistic chains than in mixed and in autistic chains, and for learners in non-autistic chains than in mixed chains only; telling participants their chain's neurotype raised rapport significantly for learners and not for teachers. For this paper the constraint is twofold: information transfer is not where the cost appeared in that task, and the rapport effects depend on role, which is why effort rather than rapport is the primary outcome here and why rapport gets a model of its own.

Rapport in groups. In 36 groups of four, all-autistic, all-non-autistic and three-to-one majorities, there was no overall group-type difference in composite rapport [29]. All-autistic groups rated their interaction more enjoyable and friendlier than both kinds of mixed group; autistic participants' ease and enjoyment fell as non-autistic members were added, the composite interaction not reaching significance, while non-autistic participants' composite rapport stayed stable; the item-level results are uncorrected for multiple comparisons and are called preliminary by their authors. Neither this study nor the chain study shows that majority-composition settings favour non-autistic rapport in general. The group study did not manipulate norms, so any reading of its item-level pattern through the cost model, the autistic participants' falling ease as the session's conventions shift toward the majority's, is a hypothesis for Study 1 and not a result.

Table 4: Robustness to departures from the generating model: 60 dyads, 300 simulated studies per row, the full model fitted. Each cell gives the bias of the estimate and, in parentheses, the coverage of its nominal 95% interval in percent; the last two columns give the power of the primary joint test and the share of studies in which the Bayesian information criterion selected a model with the norm terms, in percent. The third and the last rows generate the mismatch effects at half their values ($\lambda_s = 0.30$, $\lambda_k = 0.40$); measurement error adds noise of standard deviation 0.5 to each profile coordinate and 0.3 to each coded style and threshold coordinate, one draw per study; the correlated baselines add 0.5 per unit of partner distance and 0.4 per unit of profile extremity (the Euclidean norm of the style profile); the levels regression carries the four profile coordinates as covariates throughout; the unequal variances are 0.6 under A and 1.0 under B; the correlated predictors carry 0.6 of the centred style into competence.

Departure	γ	λ_s	λ_k	μ	Joint	BIC
As generated	−0.010 (95)	−0.002 (97)	+0.002 (96)	+0.002 (91)	100	100
Profiles and codings measured with error	−0.006 (55)	+0.017 (84)	−0.095 (74)	+0.066 (91)	100	100
Mismatch effects halved	+0.001 (94)	−0.005 (94)	+0.009 (94)	+0.020 (93)	100	100
Baselines correlated with profiles and partner distance	−0.003 (95)	−0.001 (94)	+0.003 (95)	+0.556 (9)	100	100
Unequal residual variances across conditions	+0.007 (97)	+0.004 (92)	+0.003 (95)	+0.000 (92)	100	100
Style and competence correlated within person	−0.003 (95)	+0.000 (91)	−0.009 (92)	+0.006 (93)	100	100
All five together, effects halved	+0.012 (80)	+0.004 (91)	−0.055 (89)	+0.660 (2)	100	100

Perception and the double-empathy model. Non-autistic observers form less favourable first impressions of autistic adults from brief samples, on the basis of presentation rather than content [30], which is one reason the cost of a mixed interaction cannot be read off its outcome alone. The distinct communication strategies model [8] is characterized in Section 6; it supplies the paper's strongest formal rival and the reason the empathy gap is measured.

10 Outside this paper

Three things the source raises are extensions, and the paper says of each what it would take.

Branch geometry. The Branch Principle is a hypothesis about the structure of cognitive profiles: at matched resolution, non-autistic profiles form one type, or few, and autistic profiles form several, in the strong form about as distant from each other as from the non-autistic type. Its test fits the same latent-structure models to both groups on the same cognitive-profile measures, a mixture model, an additive tree [31] and a dimensional model, and compares them by likelihood; the recent decomposition of autism's phenotypic heterogeneity into classes with distinct genetic programs [32] is evidence about clinical phenotype at a different grain, not about the cognitive profiles the principle concerns. The geometry is independent of the cost model and is not support for mutual advantage: equal distances do not imply crossed advantages. With $r = (0, 0)$, $b = (1, 1)$ and $c = (1, -1)$, both b and c lie at distance $\sqrt{2}$ from r, yet b dominates r while c and r cross.

Linking the geometry to Proposition 1's premise needs an argument of its own, and an asymmetric notion of similarity [33] may be the right instrument for it.

Linguistic masking. The source names a phenomenon a layer below behavioural camouflaging [34, 35, 36]: a speaker who learns a cognitive term's public extension case by case, over a private intension that differs from the dominant group's, so that the difference shows only when the term is extended to new cases. Its signature, a generalization gradient, is necessary for it and not unique to it, since concept knowledge, familiarity, confidence and learning history produce the same pattern; what would tie the gradient to the mechanism is a private organization of the instances, measured by similarity sorting without the term, predicting the divergence on novel instances over and above those, with an experimental model in which a private organization of a novel domain is induced before a public term is taught over a different one. Which instruments it would reach is open: an eight-item alexithymia score shows practically ignorable differential item functioning between autistic and non-autistic adults [37]. This is a second paper.

Shared absence. A further hypothesis, that non-reference branches share the absence of the reference's conventions and so find each other's interactions cheaper even where their own preferences differ, would separate rapport from the prediction of preferences. Its added premises are stated when it is tested, against familiarity, stigma, preference variability and disclosure, and it is not an advance prediction of anything reported here.

11 Discussion

What the model adds. The thought experiment of Section 3.1 shows that the deficit reading of a mixed interaction's difficulty depends on where the reference stands. The cost model places part of the difficulty in the rules, and makes the placement measurable: the share of the group gap that moves when the rules move, person by person, from profiles measured in advance. That share is what Study 1 estimates, and the model family is what keeps the estimate honest, since a general benefit of explicit rules and a mismatch with the partner are both allowed to carry the data if they can.

The conditional status of the conclusions. Nothing above establishes that effort tracks mismatch; the paper is built so that the question can be settled. The exact results, reversibility and the reversal condition, hold of the definitions and the model; the cost model is a hypothesis; the Branch Principle is an extension; balance is a conjecture. The reversal condition is a caution to the design: a study that finds the gap reduced by more than the baseline gap has not found a reversal, and a study powered for the reduction is not powered for the crossing.

Limitations the design carries.

- *Measurement.* The model's content is only as good as the commensurability of the profiles and the norm codings and the choice of d and w_n. Fixing them in advance protects the test from being fitted to its outcome, not from a poor choice: mismeasured profiles attenuate the slopes, a shared coding error biases the condition effect with no interval widening to show it (Section 8), and a metric that is not in fact common can manufacture an association as easily as hide one. The pilot's calibration is the remedy.

- *Two rule sets.* Two norms identify the norm terms but not their functional form; a third rule set, as in the simulation's withheld condition, is the natural extension and the strongest test.
- *Dyads.* The within-person test is clean, but the partner term is identified between dyads, needs 90 or more pairs at the simulated effect size, and rests on the zero-conditional-mean assumption of Section 8; a round-robin design is the one that identifies it within persons.
- *Scope.* Both conditions impose explicit instructions, so what the first study measures is adaptation to experimental protocols. Extending its result to the norms of ordinary interaction needs evidence that the protocols represent them, which a comparison of the pilot's adherence codings with recordings of unconstrained conversation can begin to supply.
- *Generality.* The groups studied are autistic and non-autistic adults, since that is where the question arose and where the literature is; the model is stated for any two groups whose communicative profiles differ, and its application elsewhere is a hypothesis in each case.
- *Participation.* The rule sets, the profile instruments and the interpretation of effort should be developed with autistic researchers and participants, not only about them; the design's choices that are left open are left open for that reason.

What would advance the account most. Not a further qualification, but the study itself, preregistered, with its third rule set; and, for the extensions, a cognitive-profile data set at matched resolution on which the branch geometry can be fitted rather than argued.

A note on provenance. The two principles, the toggle observation and the reading of a label as a partial information state are the larger work's [1], as Appendix C records; the formal apparatus, the cost model, the design and the simulation were developed in October 2026.

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A The simulation’s estimators

For participant i in dyad $d(i)$ and rule sets A and B, write $y_{i,n}$ for the effort under (5). The within-person difference and the person mean are

$$y_{i,\mathrm{B}} - y_{i,\mathrm{A}} = \gamma + \lambda_s\,\Delta d_i + \lambda_k\,\Delta \mathrm{sh}_i + \big(\varepsilon_{i,\mathrm{B}} - \varepsilon_{i,\mathrm{A}}\big), \tag{6}$$

$$\tfrac{1}{2}\big(y_{i,\mathrm{B}} + y_{i,\mathrm{A}}\big) = \alpha + \Delta a\, g_i + \tilde{a}_i + u_{d(i)} + \lambda_s\,\bar{d}_i + \lambda_k\,\overline{\mathrm{sh}}_i + \mu\,\mathrm{pd}_{d(i)} + \tfrac{1}{2}\big(\varepsilon_{i,\mathrm{B}} + \varepsilon_{i,\mathrm{A}}\big), \tag{7}$$

with Δd_i and $\Delta \mathrm{sh}_i$ the changes in premeasured style mismatch and competence shortfall from A to B, $\bar{d}_i$ and $\overline{\mathrm{sh}}_i$ their means, g_i the group indicator, $\tilde{a}_i$ the baseline's zero-mean part, α an intercept absorbing $\frac{1}{2}\gamma$, u_d the dyad effect the simulation adds to (5) and pd_d the partners' style distance; the fitted between equation also carries the four measured profile coordinates s_i, k_i as covariates, omitted from the display. The group indicator carries the baseline gap Δa; without it, where the rule sets are not mirror images of each other, the group difference in mean mismatch would be absorbed by μ and the shared λ's, and the real study's rule sets will not be mirror images.

Two assumptions make the decomposition a likelihood, and balance supplies neither. The residual variances are equal across conditions and the errors jointly normal: then

$$\mathrm{Cov}\Big(\varepsilon_{i,\mathrm{B}} - \varepsilon_{i,\mathrm{A}},\ \tfrac{1}{2}\big(\varepsilon_{i,\mathrm{B}} + \varepsilon_{i,\mathrm{A}}\big)\Big) \;=\; \tfrac{1}{2}\big(\,\mathrm{Var}\,\varepsilon_{i,\mathrm{B}} - \mathrm{Var}\,\varepsilon_{i,\mathrm{A}}\big) \;=\; 0,$$

and under normality the difference and the mean are independent, so that (6) identifies γ, λ_s and λ_k with the baseline, the dyad effect and the partner term cancelled, and (7) identifies μ and Δa; the robustness study reports what unequal variances do. Second, the dyad effect cancels from the differences but not from the means, whose two partners it correlates with covariance σ_u^2 (0.16 at the simulation's setting), so the N means are not independent observations. The N differences are independent of each other and of the means; within each dyad the two partners' means, with compound-symmetric covariance, are rotated into their sum and their difference, each divided by $\sqrt{2}$, which are uncorrelated and under normality independent, with variances $v + \sigma_u^2$ and $v - \sigma_u^2$ for v the variance of a person mean about its expectation, and the design of each rotated part is the between design rotated the same way. The likelihood is therefore a product over $2N$ independent Gaussian terms, the N differences with variance $2\sigma_\varepsilon^2$, the $N/2$ rotated sums and the $N/2$ rotated differences, each part with its own variance and the coefficients shared between the parts; the maximum-likelihood fit is iterated weighted least squares, run for forty iterations, well past convergence. Models are compared by $-2\log L + k\log(2N)$, with k the number of coefficients plus the three variances and $2N$ the number of independent terms, which is also the number of effort observations. The identification of μ rests on the zero-conditional-mean assumption of Section 8, the baselines $\tilde{a}_i$ having zero conditional mean given the regressors of (7) and the four measured profile coordinates, which the between equation carries as covariates in every model fitted.

The parameter-recovery study fits each equation by least squares with cluster-robust standard errors by dyad and tests (λ_s, λ_k) jointly by a Wald statistic on the clustered covariance. The model-recovery study compares the four restrictions by the criterion above and by the predictive log score of the held-out dyads under the fitted Gaussian parts, six folds. The withheld-rule-set study fits (6) on the training dyads, predicts $y_{i,\mathrm{C}} - y_{i,\mathrm{A}}$ for the held-out dyads from the fitted λ's and the premeasured mismatch under C, centres the prediction and the observation within the held-out set, and compares the root-mean-square error with the noise floor $\sqrt{2}\,\sigma_\varepsilon\sqrt{1 - 1/n_f}$, n_f the number of held-out participants.

B Preregistration items

The preregistration fixes, before any participant is run: the instruments for s_i and k_i and their scales; the two rule sets' texts, their coded s_n^*, θ_n and w_n, and the rater protocol; the distance d; the effort measures, subjective and behavioural, and the expected and perceived empathy reports; the practice-to-criterion rule and the adherence coding; the counterbalancing; the primary outcome, the workload score, with the repair index secondary; the primary analysis of (6) with dyad-clustered

errors, the three-part decision rule of Section 7 and how rejections with a sign contradiction or of undetermined direction are recorded, the secondary one-sided tests, and the smallest effects of interest on the fixed scales; the period and carryover terms; the separate analyses of training burden, with their estimands; the actor–partner model for levels with the profile covariates and its zero-conditional-mean assumption; the model comparison by the Bayesian information criterion on the exact likelihood; the mediation analysis with its sensitivity analysis; the three failure conditions of Section 7; and the sample size, 60 mixed dyads if the primary prediction alone is at issue, 90 if the full family is to be compared, and 120 if the equivalence tests at the smallest effects of interest are to have 80% power or more, at the simulated effect sizes, which are to be replaced by pilot estimates where available. Before the preregistration, a pilot establishes whether the profiles and the norm requirements can be scored on shared scales, whether participants can follow both rule sets to criterion and at what cost in training, whether the effort measures track the intended construct against performance and adherence, and the reliabilities of the empathy reports and of the rule-set codings.

C The source passages

The passages below are quoted as written in the larger work [1], from the section "The Aethic Mindset and Autistic Savants" and its subsection "Illustrating the Bias of Perspective in Diagnosing Logic", and from the paragraph "The decisive fingerprint: it toggles" of the section "Aethic Reasoning Against the Modal Structure of Human Language". The source dates the two principles to a short video the author recorded on June 8, 2024, at 2:39 PM ET.

The Branch Principle, and its motivation.

> "Ontologically neurodivergent people are better understood as a collection of categorical branches and 'allistic' is a single branch, such that two random branches are typically as different from one another as they each are to a third."

> "A better model for understanding autistic minds than the autistic-allistic binary is more of a tree model, such that allistic people are one branch alone and the rest have been all categorized as 'autistic'. Inadvertently speaking, then, this proposes that an autism diagnosis is not actually diagnosing a person's traits to the level of specificity of an allistic person (as would assign them a particular branch) but instead broadly classifies them as 'non-allistic' for their lacking allistic traits. In reality, though, each branch comes with a different set of cognitive intuitions, which are separate enough so that two autistic people drawn from two different branches tend to be as different from one another at the cognitive level as either of them are from allistic people."

> "This is meant to be directly analogous to the post-Darwinian tree of life model treating humans as a branch, as opposed to the pre-Darwinian model in which humans and animals are two categories."

The Principle of Symmetry, and its intended content.

> "You can always redraw the conventional category boundaries in psychology so that allistic and a given neurodivergent branch 'trade places phenomenologically', in the sense that the exact behavioral diagnosis, as an Aethus, is made to reverse isomorphically."

"As an Aethus" is glossed at the source as "a partial information state about that behavioral diagnosis rather than an exhaustive description". The principle's intended content is given in the same passage with the author's own case as the instance, offered in this paper as motivation only:

> "What's fundamentally wrong about this is that it fundamentally misunderstands what's actually going on: what's going on is utter, complete symmetry between cognitive branches. That is, for allistic people versus a given branch of autistic people, there will be roughly half-and-half instances across traits for which the autistic person is infantile versus for which the allistic person is infantile. Using myself as an example, at age twenty I still had the emotional maturity of a thirteen year old, but at age thirteen I already had the mathematical maturity of at least a twenty year old. Had my branch been the reference instead, allistic people would look just as profoundly infantile in so many different ways to how I looked in social, emotional, and life skills."

The random-partition illustration.

> "This is also where categorizing all the non-allistic branches as one 'autism' is measurably problematic: you can find an autistic branch for which any trait is infantile, thereby seeming to confirm your prior that 'look, autistic people are infantile in this way'. Running the same logic on a random partition where we put half the branches, including the allistic one, in one category while the rest are placed in the other, and we could just as easily misinterpret allistic people as infantile in any conceivable way by finding it in another branch of the categorization and then mistakenly importing it to an assessment of allistic people."

The 2024 version of the thought experiment, on "logical" and "nuanced". The passage that follows the two principles:

> "The observation, now, is in how mathematics itself, especially in an educational setting, is commonly called 'logical' in the sense of there only being right and wrong answers, whereas the humanities are instead called 'nuanced' in the sense that they resist binary 'rightness' of answers. What I want to point out now is that this take itself is actually biased to the allistic perspective, which has some intriguing philosophical implications once we start to unpack it. First of all, notice how due to the second bullet point, it is actually universal that anyone learning the most introductory material to a topic of a different intuitive landscape will report the same characterization. In order for a child to learn literature, they must first start by learning the alphabet and the sounds they make. In this sense, like twelfth grade math, kindergarten literature has clear rules—either the letter is written this way or it is not, and either the letter makes this sound or it does not. The fact that kindergarten literature is at an isomorphic introductory level as twelfth grade mathematics is then simply an illustration of how the general population—most of them being of the allistic cognitive branch—hold a vastly higher intuitive capacity for literature over mathematics. So in this sense comparing twelfth grade mathematics to twelfth grade literature is the wrong reference class, hence contributing to this bias of perspective that literature is inherently more nuanced than math."

The seat of the cost. From a footnote of the same section:

> "[N]otice how many autistic people themselves are described as taking a script to allistic conversations. Once again, notice how if the shoe were on the other foot, for any given non-allistic cognition branch allistic people practicing its topics would be interpretable as using a script."

The toggle.

> "One further fingerprint is of a different kind, and it is decisive against the strongest rival reading: that the convention is not a convention at all but a fixed cognitive limitation. A limitation does not toggle. The switching just described—an adult moving, on demand, between the attached picture and the conventional one within a single day, by context—is a *dissociability* result: the convention is a mode the platform runs, selectable, not a ceiling the platform imposes."